MOBILE APP DEVELOPMENT PROGRAM

Mobile App Development

Full Program — Course Syllabus

Build native-performing iOS and Android apps using Flutter, Dart, React Native, Firebase, and offline-first databases.

Core Technologies Mastered

Dart · Flutter · React Native · Firebase · REST/GraphQL APIs · SQLite / Offline-First Storage · CI/CD

Program Overview

This 14–16 week, project-driven curriculum takes learners from Dart and Flutter fundamentals through state management, backend and cloud integration, native device capabilities, and the React Native/Expo ecosystem, culminating in a fully tested app shipped through CI/CD to the App Store and Google Play. Each module combines theory, hands-on exercises, and a mini-project, culminating in a portfolio-ready capstone app.

No	Module	Duration	Focus Area
1	Dart & Flutter Fundamentals	Week 1 – 3	Flutter
2	State Management & Architecture	Week 4 – 5	Flutter
3	Backend & Cloud Integrations	Week 6 – 8	Firebase / APIs
4	Device Features & Native APIs	Week 9 – 10	Native / Hardware
5	React Native & Expo Ecosystem	Week 11 – 13	React Native
6	Testing, CI/CD & App Store Launch	Week 14 – 16	DevOps / Release

Module 1: Dart & Flutter Fundamentals

Duration	Week 1 – 3
Outcome	Ability to build and style multi-screen Flutter apps using Dart and core widgets.

Topics Covered

- Dart fundamentals: variables, types, null safety, functions, classes
- Asynchronous Dart: Futures, async/await, Streams
- Flutter architecture: widget tree, element tree, render tree
- Stateless vs. Stateful widgets and the widget lifecycle
- Layout widgets: Row, Column, Stack, ListView, GridView
- Navigation and routing (Navigator 1.0/2.0, named routes)
- Styling: themes, custom fonts, responsive layouts
- Forms and input validation

Mini-Project: Build a multi-screen Flutter app (e.g., recipe browser or expense tracker) with navigation and forms.

Module 2: State Management & Architecture

Duration	Week 4 – 5
Outcome	Skill to structure scalable Flutter apps using modern state management and architecture patterns.

Topics Covered

- setState limitations and when to reach for a state manager
- Provider and InheritedWidget fundamentals
- Riverpod: providers, notifiers, and dependency injection
- BLoC pattern: events, states, and streams
- App architecture: MVVM and clean architecture in Flutter
- Separating UI, business logic, and data layers
- Dependency injection with get_it
- Unit testing state and business logic

Mini-Project: Refactor the Module 1 app into a layered architecture using Riverpod or BLoC.

Module 3: Backend & Cloud Integrations

Duration	Week 6 – 8
Outcome	Ability to connect mobile apps to authenticated backends, real-time data, and offline-first storage.

Topics Covered

- REST API integration with Dio/http and error handling
- GraphQL basics for mobile clients
- Firebase Authentication (email/password, OAuth, phone)
- Cloud Firestore: real-time data, queries, and security rules
- Firebase Storage for media uploads
- Push notifications with Firebase Cloud Messaging
- Offline-first design with SQLite/Drift and local caching
- Syncing local and remote data, conflict resolution basics

Mini-Project: Add authentication, real-time data sync, and offline support to the app (e.g., a chat or task-sharing app).

Module 4: Device Features & Native APIs

Duration	Week 9 – 10
Outcome	Capability to integrate native device hardware and platform-specific functionality.

Topics Covered

- Camera and photo library access
- Geolocation and maps integration
- Local notifications and background tasks
- Sensors: accelerometer, gyroscope basics
- Permissions handling across iOS and Android
- Platform channels: calling native iOS/Android code from Flutter
- Biometric authentication (Face ID/Touch ID, fingerprint)
- App performance profiling and memory management

Mini-Project: Extend the app with camera capture, location tagging, and push/local notifications.

Module 5: React Native & Expo Ecosystem

Duration	Week 11 – 13
Outcome	Working proficiency building cross-platform apps with React Native and the Expo toolchain.

Topics Covered

- React Native fundamentals: components, JSX, props, state, hooks
- Expo managed workflow vs. bare workflow
- Navigation with React Navigation
- Styling with StyleSheet and Flexbox layouts
- State management with Context API and Redux Toolkit
- Consuming REST APIs and handling async data with React Query
- Using native modules and Expo SDK APIs (camera, location, notifications)
- Comparing Flutter and React Native: trade-offs and when to choose each

Mini-Project: Rebuild a core app flow in React Native/Expo to compare performance and developer experience.

Module 6: Testing, CI/CD & App Store Launch

Duration	Week 14 – 16
Outcome	End-to-end capability to test, automate, and ship a production mobile app.

Topics Covered

- Unit and widget testing in Flutter; component testing in React Native
- Integration and end-to-end testing (integration_test, Detox)
- Crash reporting and analytics (Firebase Crashlytics/Analytics)
- Setting up CI/CD pipelines (GitHub Actions, Codemagic, EAS Build)
- App signing, provisioning profiles, and build variants
- Preparing app store listings, screenshots, and metadata
- Submitting to the Apple App Store and Google Play Store
- Post-launch monitoring, versioning, and staged rollouts

Mini-Project: Capstone — test, containerize a CI/CD pipeline for, and publish the final app to both app stores (or a staged/internal track).

Suggested Tools & Stack

Category	Tools
Languages	Dart, JavaScript/TypeScript
Frameworks	Flutter, React Native, Expo
Backend / Cloud	Firebase (Auth, Firestore, Storage, FCM), REST/GraphQL APIs
Local Storage	SQLite, Drift, AsyncStorage
Testing	flutter_test, integration_test, Jest, Detox
CI/CD & Release	GitHub Actions, Codemagic, EAS Build, Fastlane

Assessment & Certification

- Weekly hands-on coding assignments and exercises aligned to each module
- Mini-project after every module (Modules 1–6)
- Mid-program checkpoint: end-to-end Flutter app combining state management and backend integration (Modules 1–3)
- Capstone project: full cross-platform app tested, automated, and published to app stores
- Final proficiency evaluation covering architecture, native integration, and release engineering

Upon successful completion, participants will demonstrate production-level mobile engineering proficiency across the full app lifecycle — from Flutter and React Native development through state management, backend integration, native device features, testing, and app store deployment.

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